Today

- Free particle: Wave packets, phase and group velocities
- Begin topic 3
  - Axiomatic principles of quantum mechanics

2.10 Free particle

Consider a particle of mass m and energy E that is not confined by a potential, or $E > V(x)$ for all x. We will be primarily focusing on the special case of a completely free particle with no forces acting on it,

$$V(x) = 0, \quad -\infty < x < \infty. \tag{2.10.1}$$

First, classically, for $V(x) = 0$, Newton's second law of motion gives $m\ddot{x} = 0$, and taking the initial position of the particle at $t = 0$ to be at the origin, we have

$$x(t) = v_{cl}t, \tag{2.10.2}$$

where v_{cl} is the classical velocity of the particle.

In quantum mechanics, for $V(x) = 0$, the time-independent Schrödinger equation is

$$\psi''(x) + k^2\psi(x) = 0, \qquad k \equiv \sqrt{\frac{2mE}{\hbar^2}}, \tag{2.10.3}$$

The time-dependent general solution is

$$\psi(x,t) = Ae^{ikx - i\frac{\hbar k^2}{2m}t} + Be^{-ikx - i\frac{\hbar k^2}{2m}t} = Ae^{i(kx-\omega t)} + Be^{-i(kx+\omega t)} = Ae^{ik(x-v_p t)} + Be^{-ik(x+v_p t)}, \tag{2.10.4}$$

where

$$\omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m} \tag{2.10.5}$$

is the angular velocity,

$$v_p = \frac{\omega}{k} = \frac{\hbar k}{2m} = \frac{p}{2m} = \frac{1}{2}v_{cl} \tag{2.10.6}$$

is the phase velocity, p is the momentum of the particle, and v_{cl} is the classical velocity of the particle. Note that the phase velocity is half the classical velocity of the particle. The first term in (2.10.4) corresponds to a wave propagating the right, and the second term a wave propagating

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to the left.

Focusing on a free particle traveling to the right, we have

$$\psi(x,t) = Ae^{ikx - i\frac{\hbar k^2}{2m}t}. \tag{2.10.7}$$

Because $|\psi(x,t)|^2 = |A|^2$, $\int_{-\infty}^{\infty} |\psi(x,t)|^2 dx = \int_{-\infty}^{\infty} |A|^2 dx = \infty$, and the wavefunction is not normalizable.

> **Question 2.10.1:** A physical free particle (such as a free electron), (A). can be described by a stationary state. (B). has definite position. (C). should have a wavefunction that is mostly concentrated in some finite region in space.

The question is how we could write a wavefunction that is peaked around a finite region in space and moves along with the particle. In particular, we would like to write a wavefunction that looks like

$$\psi(x,t) = A(x,t)e^{ik_0 x - i\omega_0 t}, \tag{2.10.8}$$

where k_0 is the peak value of the wave vector and $A(x,t)$ is the amplitude at point x at time t, which we would like to be concentrated around a finite region in space, where the particle is most likely to be found.

We can construct a wavefunction of desired profile by adding plane waves of different wavelengths and amplitudes. Suppose we integrate over continuum values of wave vectors and write

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk\, \phi(k) e^{ikx - i\omega(k)t}, \tag{2.10.9}$$

where the prefactor of $\frac{1}{\sqrt{2\pi}}$ is a choice for a convention of notation as we point out below. Here, $\phi(k)$ is the amplitude for the plane wave of wave vector k. Because $\phi(k)$ is a stationary state of definite energy $E = \frac{\hbar^2 k^2}{2m}$, its time evolution follows immediately,

$$\phi(k,t) = \phi(k) e^{-i\omega(k)t}, \tag{2.10.10}$$

and we can rewrite

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk\, \phi(k,t) e^{ikx}, \tag{2.10.11}$$

which is what is called the Fourier transform. In particular, at time $t = 0$, $\psi(x) = \psi(x,0)$, we write

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk\, \phi(k) e^{ikx}, \tag{2.10.12}$$

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Conversely, we can write $\phi(k,t)$ in terms of a space integral over $\psi(x,t)$ as

$$\phi(k,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, \psi(x,t) e^{-ikx}, \tag{2.10.13}$$

which is what is called the inverse Fourier transform. At $t = 0$, with $\phi(x) = \phi(x,0)$,

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, \psi(x) e^{-ikx}, \tag{2.10.14}$$

If we put (2.10.13) back in (2.10.12), we have

$$\begin{aligned} \psi(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx'\, \psi(x') e^{-ikx'} \right) e^{ikx} \\ &= \int_{-\infty}^{\infty} dx' \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} dk\, e^{-ik(x'-x)} \right) \psi(x') \\ &= \int_{-\infty}^{\infty} dx'\, \delta(x'-x) \psi(x') \\ &= \psi(x), \end{aligned} \tag{2.10.15}$$

where

$$\delta(x'-x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk\, e^{-ik(x'-x)} \tag{2.10.16}$$

is the delta function. The functions $\psi(x)$ and $\phi(k)$ are called Fourier transforms of each other, and the corresponding expressions of one in terms of the other are Fourier transforms. Note that the factor of $\frac{1}{\sqrt{2\pi}}$ appears symmetrically in both, and that was the reason for the convention of introducing it in (2.10.9) first. The delta function arises because, when we add the plane waves of equal amplitude over all wave vectors, and $x' - x \neq 0$, then the plane waves destructively interfere and completely cancel out. On the other hand, for $x' - x = 0$, the exponent vanishes and we are integrating over a constant, which results in infinity.

For a wave vector amplitude $\phi(k)$ that is peaked at a particular frequency corresponding to wave vector $k = k_0$, we can make a Taylor series expansion of $\omega(k)$ about k_0 and keep only the first few terms, since only frequencies around $k = k_0$ are available because of the profile of $\phi(k)$,

$$\omega(k) = \omega_0 + \frac{d\omega(k)}{dk}\Big|_{k=k_0} (k - k_0) + \frac{1}{2} \frac{d^2\omega(k)}{dk^2}\Big|_{k=k_0} (k - k_0)^2 + \ldots, \tag{2.10.17}$$

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where $\omega_0 = \omega(k_0)$. Using (2.10.17) in the exponent of the integrand in (2.10.9),

$$\begin{aligned} kx - \omega(k)t &= (k_0 + k - k_0)x - (\omega_0 + \frac{d\omega(k)}{dk}|_{k=k_0}(k-k_0) + \frac{1}{2}\frac{d^2\omega(k)}{dk^2}|_{k=k_0}(k-k_0)^2 + ...)t \\ &= k_0\left(x - \frac{\omega_0}{k_0}t\right) + (k-k_0)\left(x - \frac{d\omega(k)}{dk}|_{k=k_0}t\right) - \frac{1}{2}\frac{d^2\omega(k)}{dk^2}|_{k=k_0}(k-k_0)^2 t + ... \\ &= k_0\left(x - v_p t\right) + (k-k_0)\left(x - v_g t\right) - \frac{1}{2}a^2(k-k_0)^2 t + ..., \end{aligned} \tag{2.10.18}$$

where

$$v_p = \frac{\omega_0}{k_0} \tag{2.10.19}$$

is the phase velocity,

$$v_g = \frac{d\omega(k)}{dk}|_{k=k_0} \tag{2.10.20}$$

is called the group velocity, and

$$a^2 = \frac{d^2\omega(k)}{dk^2}|_{k=k_0}. \tag{2.10.21}$$

If we use the $\omega(k)$ given by (2.10.5), we have

$$v_p = \frac{\omega_0}{k_0}, \qquad v_g = \frac{\hbar k_0}{m} = v_{cl}. \tag{2.10.22}$$

Using (2.10.18) in (2.10.9), and keeping only terms up to quadratic in the Taylor expansion of $\omega(k)$, we have

$$\begin{aligned} \psi(x,t) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dk\, \phi(k) e^{ik_0(x-v_p t)} e^{i(k-k_0)(x-v_g t) - \frac{1}{2}a^2(k-k_0)^2 t} \\ &= \frac{1}{\sqrt{2\pi}} e^{ik_0(x-v_p t)} \int_{-\infty}^{\infty} dk\, \phi(k) e^{-\frac{1}{2}a^2(k-k_0)^2 t + i(k-k_0)(x-v_g t)} \\ &= \frac{1}{\sqrt{2\pi}} e^{ik_0(x-v_p t)} \int_{-\infty}^{\infty} dk\, \phi(k) e^{-\frac{1}{2}a^2\left[(k-k_0) - i\frac{x-v_g t}{a^2}\right]^2} e^{-\frac{(x-v_g t)^2}{2a^2}} \\ &= \frac{1}{\sqrt{2\pi}} e^{ik_0(x-v_p t)} e^{-\frac{(x-v_g t)^2}{2a^2}} \int_{-\infty}^{\infty} dk\, \phi(k) e^{-\frac{1}{2}a^2\left[(k-k_0) - i\frac{x-v_g t}{a^2}\right]^2} \end{aligned} \tag{2.10.23}$$

Thus, the group velocity is the classical velocity at which the particle travels, since the center of the modulating Gaussian function $e^{-\frac{(x-v_g t)^2}{2a^2}}$ moves with velocity v_g. On the other hand, the plane wave of peak wave vector k_0 inside the packet propagates at the phase velocity v_p. The wavefunction $\psi(x,t)$ represents a wave packet for a given choice of $\phi(k)$ peaked around some $k = k_0$.

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The general formula we use for finding $\psi(x,t)$ given $\phi(k,t)$ is the Fourier transform (2.10.11), and the general formula for finding $\phi(k,t)$ given $\psi(x,t)$ is the inverse Fourier transform (2.10.13), which can be applied for a particle with appropriate $\omega(k)$ even when the potential $V(x) < E$ is not necessarily zero. For a free particle with $V(x) = 0$, we have $E = \hbar\omega = \frac{\hbar^2 k^2}{2m}$, or $\omega(k) = \frac{\hbar^2 k^2}{2m}$, and we can use this in (2.10.11) and (2.10.13) to determine $\psi(x,t)$ and $\phi(k,t)$. In this case, the Taylor expansion about a k_0 is exhausted at second order, and the Taylor expansion that resulted in (2.10.23) produces the exact answer. However, for a more general $\omega(k)$, with forces acting on the particle, (2.10.23) is a good approximation only when $\phi(k)$ is sharply peaked and appropriately narrow around $k = k_0$.

Question 2.10.2: Discuss qualitatively why is the group velocity could be related to the derivative of $\omega(k)$ with respect to k at all?

Question 2.10.3: Discuss why the wavelength of the wave packet is larger than the wavelength of the plane wave at peak frequency using (2.10.23).

2.10.1 Rectangular wave vector amplitude

Now consider a choice of rectangular shape for the amplitude $\phi(k)$,

$$\phi(k) = \begin{cases} \frac{1}{\sqrt{\Delta k}}, & -\frac{\Delta k}{2} < k < \frac{\Delta k}{2} \\ 0, & |k| > \frac{\Delta k}{2}. \end{cases}$$

This is shown in the figure below.

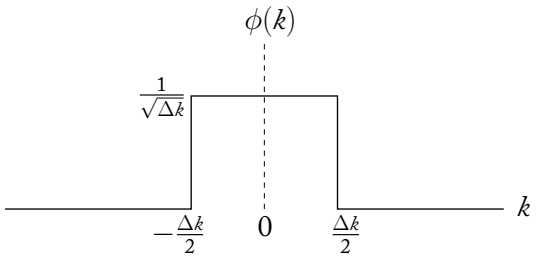


Figure 2.10.1: Rectangular wave vector amplitude.

Note that $\phi(k)$ is normalized, $\int_{-\infty}^{\infty} |\phi(k)|^2 dk = \frac{1}{\Delta k}\Delta k = 1$. Using (2.10.28) in (2.10.9),

$$\psi(x,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk\, \phi(k) e^{ikx} = \frac{1}{\sqrt{2\pi}} \int_{-\frac{\Delta k}{2}}^{\frac{\Delta k}{2}} dk\, \frac{1}{\sqrt{\Delta k}} e^{ikx}$$

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$$= \frac{1}{\sqrt{2\pi\Delta k}}\frac{1}{ix}\left(e^{i\frac{\Delta k}{2}x} - e^{-i\frac{\Delta k}{2}x}\right) = \frac{1}{\sqrt{2\pi\Delta k}}\frac{2}{x}\sin\left(\frac{\Delta k}{2}x\right). \tag{2.10.24}$$

A schematic plot of $\psi(x,0)$ is shown in the figure below.

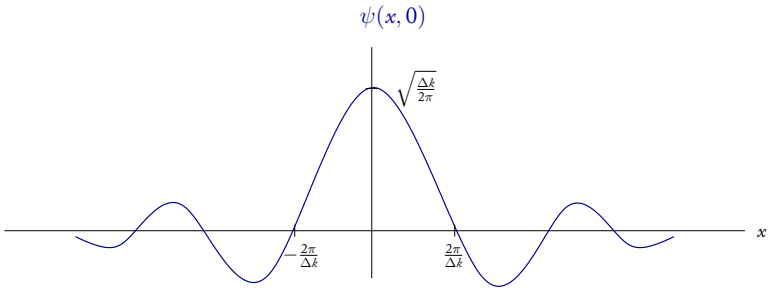


Figure 2.10.2: Wave packet for a rectangular wave vector amplitude.

Thus, we have a wave packet that is concentrated at the origin and falls off at large x, which is a more appropriate description for a particle that is located around the origin. We also observe that the width of the central peak of the wave packet Δx, where the particle is most likely to be found, can be characterized by the width of the range of the nodes at $\pm\frac{2\pi}{\Delta k}$ on the $x-$axis,

$$\Delta x = \frac{4\pi}{\Delta k}. \tag{2.10.25}$$

The nodes of the wavefunction occur when $\sin\left(\frac{\Delta k}{2}x\right) = 0$, or $\frac{\Delta k}{2}x = n\pi$, $n = \pm 1, \pm 2,...$, and the width of the central peak follows from $n = \pm 1$. On other hand, the width in momentum is $\Delta p = \hbar\Delta k$. Therefore,

$$\Delta x \Delta p = 4\pi\hbar. \tag{2.10.26}$$

We note that the smaller the width in momentum, the larger the width in position. In other words, the more certain the value of the momentum, the less certain the position of the particle, consistent with the uncertainty principle.

Moreover, if Δk is small such that $\Delta k x << 1$, then

$$\psi(x,0) = \frac{1}{\sqrt{2\pi\Delta k}}\frac{2}{x}\sin\left(\frac{\Delta k}{2}x\right) \approx \frac{1}{\sqrt{2\pi\Delta k}}\frac{2}{x}\frac{\Delta k}{2}x = \sqrt{\frac{\Delta k}{2\pi}}, \tag{2.10.27}$$

which is a constant independent of x. Therefore, when Δk or $\Delta p = \hbar\Delta k$ is small and the momentum of the particle is more precisely known, $\psi(x,0)$ is a constant and the position of the

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particle is very much uncertain. On the other hand, when Δk is large and the momentum is uncertain, the $\frac{1}{x}$ factor cannot be canceled by the x in the oscillating sin function and $\psi(x,0)$ oscillates fast with small width for the central peak, and the position of the particle is more precisely known.

2.10.2 Rectangular pulse

Now consider a choice of rectangular pulse for the initial wavefunction $\psi(x,0) = \psi(x)$,

$$\psi(x) = \begin{cases} \frac{1}{\sqrt{\Delta x}}, & -\frac{\Delta x}{2} < x < \frac{\Delta x}{2} \\ 0, & |x| > \frac{\Delta x}{2}. \end{cases}$$

This is shown in the figure below

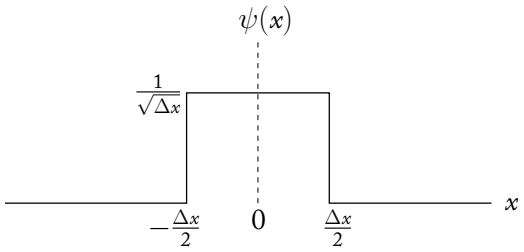


Figure 2.10.3: Rectangular pulse.

Again, $\psi(x)$ is normalized, $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \frac{1}{\Delta x}\Delta x = 1$. Using (2.10.28) in (2.10.9),

$$\begin{aligned} \phi(k,0) = \phi(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, \psi(x) e^{-ikx} = \frac{1}{\sqrt{2\pi}} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} dx\, \frac{1}{\sqrt{\Delta x}} e^{-ikx} \\ &= \frac{1}{\sqrt{2\pi\Delta x}} \frac{1}{ik} \left(e^{ik\frac{\Delta x}{2}} - e^{-ik\frac{\Delta x}{2}} \right) = \frac{1}{\sqrt{2\pi\Delta x}} \frac{2}{k} \sin\left(k\frac{\Delta x}{2} \right). \end{aligned} \tag{2.10.28}$$

A schematic plot of $\phi(k,0)$ is shown in the figure below.

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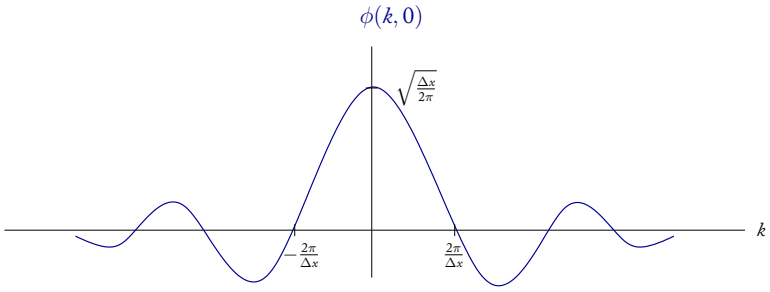


Figure 2.10.4: Wave packet in $k-$space for a rectangular pulse.

Now we have a wave packet that is concentrated at the origin in $k-$space. Its width is

$$\Delta k = \frac{4\pi}{\Delta x}. \tag{2.10.29}$$

Again, $\Delta x \Delta p = 4\pi\hbar$.

Because $\phi(k)$ is a state of definite energy $E = \frac{\hbar^2 k^2}{2m}$, we can immediately write

$$\phi(k,t) = \phi(k) e^{-\frac{\hbar k^2}{2m} t} = \frac{1}{\sqrt{2\pi\Delta x}} \frac{2}{k} \sin\left(k\frac{\Delta x}{2}\right) e^{-i\frac{\hbar k^2}{2m} t}. \tag{2.10.30}$$

With this and (2.10.11),

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk \, \frac{1}{\sqrt{2\pi\Delta x}} \frac{2}{k} \sin\left(k\frac{\Delta x}{2}\right) e^{ikx - i\frac{\hbar k^2}{2m} t}, \tag{2.10.31}$$

Again, the more precisely the position is known, the more uncertain the momentum is and vice versa. The above integral can be done, but it is unnecessarily complicated for our purpose here. It turns out that the smaller the Δx, the faster the rate the packet $\psi(x,t)$ spreads out in time. In other words, a narrower wave packet spreads out at a faster rate than a wider wave packet.

Question 2.10.4: Discuss why the improper wavefunctions are valuable.

Exercise 2.10.1: Consider a particle represented by the Gaussian wave packet $\psi(x,0) = \mathcal{A} e^{-\frac{x^2}{2a^2}} e^{ik_0 x}$ in position space at time $t = 0$, where a and k_0 are constants.

1. Normalize $\psi(x,0)$.
2. Determine the wavefunction $\phi(k,0)$ at time $t = 0$ in momentum space using the Fourier transform, $\phi(k,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \, \psi(x,0) e^{-ikx}$.
3. Check if $\phi(k,0)$ is normalized through the normalization of $\psi(x,0)$ or if we

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need to normalize again.

4. Determine $\psi(x, t)$ and show that $|\psi(x, t)|^2$ can be written as $|\psi(x, t)|^2 = \frac{1}{\sqrt{\pi}a(t)} e^{-\frac{(x-\frac{\hbar k_0}{m}t)^2}{a(t)^2}}$, and determine $a(t)$.
5. Discuss how $|\psi(x, t)|^2$ evolves in time. How long does it take for the width to change appreciably? How far has the particle traveled during this time?
6. Determine the uncertainty in the position of the particle Δx at time t.
7. Determine the uncertainty in the momentum of the particle Δp at time t.
8. Write down $\Delta x \Delta p$ at time t and check if it obeys the uncertainty principle. What is the value of $\Delta x \Delta p$ at time $t = 0$ and what happens as t increases and $t \to \infty$?

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3 Principles of quantum mechanics

Now we organize the principles of quantum mechanics more formally and axiomatically, and investigate their consequences. At the same time, we introduce and use the Dirac bra-ket notation that becomes convenient as we go along.

3.1 Axiomatic principles

1. **State vector axiom:** The state of a physical system is represented by a normalized state vector in Hilbert space. The state vector contains all information about the system. [We will represent the state vector at time t by the Dirac ket notation $|\psi(t)\rangle$, and we call it ket $\psi(t)$. We will often simply write the state as $|\psi\rangle$ with the time dependence implicitly understood.]

2. **Probability axiom:** Given a system in state $|\psi\rangle$, a measurement will find it in state $|\phi\rangle$ with probability amplitude $\langle\phi|\psi\rangle$. [We call $\langle\psi|$ a bra state, and call it bra ψ.]

3. **Hermitian operator axiom:** A physical observable (or any measurable physical quantity) is represented by a linear and Hermitian operator.

4. **Measurement axiom:**

   (a) A measurement of a physical observable results in an eigenvalue of the observable. [Denoting an observable by $\hat{A}$, we have $\hat{A}|a\rangle = a|a\rangle$, where a is the eigenvalue of $\hat{A}$ and $|a\rangle$ is the corresponding eigenvector.]

   (b) A measurement of the physical quantity represented by $\hat{A}$ collapses the state $|\psi\rangle$ before measurement into an eigenstate $|a\rangle$ of $\hat{A}$.

5. **Time evolution axiom:** The state vector evolves according to the Schrödinger equation, $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$, where $\hat{H}$ is the Hamiltonian operator for total energy.

3.2 Vector space

The state of a system is represented by a (complex) vector $|\psi\rangle$ in Hilbert space. Let us use the simple Harmonic oscillator as an example for illustration. When we studied the simple harmonic oscillator, we wrote the Schrödinger equation as $i\hbar\frac{\partial}{\partial t}\psi(x,t) = \hat{H}\psi(x,t)$, where $\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx} + \frac{1}{2}m\omega^2x^2$. Recall that the wavefunctions were functions of position and time, and given by $\psi_n(x,t)$ with corresponding energies E_n, where $n = 0, 1, 2, ...$ Why didn't we construct

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wavefunctions that are functions of momentum and time (in stead of position and time)? Indeed, it is possible to solve the Schrödinger equation of the simple Harmonic oscillator with exactly the same level of mathematics in momentum space as in position space. For some Hamiltonians, the Schrödinger equation is simpler to solve in momentum space. More generally, we do not want to be restricted to a particular space. Therefore, we write the Schrödinger equation as

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle \tag{3.2.1}$$

without a consideration of position or momentum.

In other words, the state vector is neither in position nor in momentum space. As an example, for the simple Harmonic oscillator, we have states that correspond to $n = 0$, $n = 1$, $n = 2$, ..., $n = \infty$. The Hilbert space is the space spanned by basis vectors $|0\rangle$ for the ground state, $|1\rangle$ for the ground state, etc. We can view $|0\rangle$ as a column vector of infinite number of components with 1 in the first entry and 0 for the rest. Similarly, $|1\rangle$ could be viewed as a column vector of infinite number of components with 1 in the second entry and 0 for the rest. In other words, we may view the basis vectors as

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ \vdots \end{pmatrix}, \qquad |n\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \qquad \cdots, \tag{3.2.2}$$

where the 1 in $|n\rangle$ appears in the nth entry. We denote a state by the ket $|n\rangle$ that corresponds to the energy level $E_n = (n + \frac{1}{2})\hbar\omega$, again without any consideration of position or momentum. Because, n runs from 0 to ∞ in discrete integers, we have an infinite dimensional discrete vector space. Because the time evolution produces a complex vector, even if we start with a real one, $|n(t)\rangle$ is a complex vector.

However, the vector spaces we will investigate also involve vectors with continuous elements as in the position of a particle which involves continuous and infinite number of points on a line. We will see how we describe such vectors later as we go along.

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3.2.1 Linearity

Because the Schödinger equation is linear, given two states $|\psi_1(t)\rangle$ and $|\psi_2(t)\rangle$, the linear superposition $|\psi(t)\rangle = c_1|\psi_1(t)\rangle + c_2|\psi_2(t)\rangle$, where c_1 and c_2 are complex constants, is also a solution.

Because, the state vector needs to be normalized, multiplying a state vector by an overall complex constant does not change its content. In other words, $|\psi(t)\rangle$ and $c|\psi(t)\rangle$, where c is a complex constant, are physically equivalent.

3.2.2 Properties of a vector space

The following are properties of vector spaces under addition and multiplication by complex constants.

1. Addition:
   (a) Commutativity of addition: $|\psi_1\rangle + |\psi_2\rangle = |\psi_2\rangle + |\psi_1\rangle$.
   (b) Associativity of addition: $(|\psi_1\rangle + |\psi_2\rangle) + |\psi_3\rangle = |\psi_1\rangle + (|\psi_2\rangle) + |\psi_3\rangle$.
   (c) Existence of a zero vector $|0\rangle$ which we also write as 0: There exists a vector $|0\rangle$ such that $|\psi\rangle + |0\rangle = |\psi\rangle$.
   (d) Existence of inverse $|-\psi\rangle$: There exists a vector $|-\psi\rangle$ such that $|\psi\rangle + |-\psi\rangle = |0\rangle$.
2. Multiplication by complex numbers:
   (a) Unity: Multiplying a vector by one reproduces the same vector, $1|\psi\rangle = |\psi\rangle$.
   (b) Associativity: Given two complex numbers c_1 and c_2, $(c_1 c_2)\,|\psi\rangle = c_1\,(c_2|\psi\rangle)$.
   (c) Distributivity: Given a complex number c and two vectors $|\psi_1\rangle$ and $|\psi_2\rangle$, $c\,(|\psi_2\rangle + |\psi_1\rangle)$ $c|\psi_1\rangle + c|\psi_2\rangle$.

3.2.3 Dual vector space

Given a vector space in which vectors $c|\psi\rangle$ lives, there exist a dual vector space in which $c|\psi\rangle$ is mapped to $c^*\langle\psi|$, where c^* is the complex conjugate of c and $c^*\langle\psi|$. In other words, for every ket vector in the vector space, there is a bra vector in the dual vector space. The bra state $c^*\langle\psi|$ is obtained by transposing and complex conjugating the ket state $c|\psi\rangle$. In other words, given a

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vector,

$$|\psi\rangle = \begin{pmatrix} \vdots \\ \alpha \\ \vdots \end{pmatrix}, \tag{3.2.3}$$

the dual vector is

$$\langle\psi| = \begin{pmatrix} \cdots, & \alpha^*, & \cdots \end{pmatrix}. \tag{3.2.4}$$

In particular for the basis vectors written in (3.2.2), the dual basis vectors are

$$\begin{aligned} \langle 0| = \begin{pmatrix} 1, & 0, & \cdots, & \cdots, & \cdots \end{pmatrix}, \qquad \langle 1| = \begin{pmatrix} 0, & 1, & 0, & \cdots, & \cdots \end{pmatrix}, \\ \cdots, \qquad \langle n| = \begin{pmatrix} 0, & \cdots, & 1, & 0, & \cdots \end{pmatrix}, \qquad \cdots \end{aligned} \tag{3.2.5}$$

3.2.4 Inner product

A bra $\langle\phi|$ and a ket $|\psi\rangle$ produce a complex number through the inner product (or a bra-ket) $\langle\phi|\psi\rangle = c$, where c is a complex number. The inner product has the following properties.

1. $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$. Therefore, $\langle\psi|\psi\rangle$ is real.
2. $\langle\psi|\psi\rangle$ is positive and finite for a normalizable ket vector. Therefore, we can multiply a given ket by a constant and choose $\langle\psi|\psi\rangle = 1$. We have $\langle\phi|\psi\rangle = 0$ for orthogonal kets $|\psi\rangle$ and $|\phi\rangle$. For an orthonormal set of kets, $\langle\psi_m|\psi_n\rangle = \delta_{mn}$.

3.3 Operators

An operator $\hat{A}$ acting on a state vector $|\psi\rangle$ transforms it into another state vector $|\phi\rangle$,

$$\hat{A}|\psi\rangle = |\phi\rangle. \tag{3.3.1}$$

Question 3.3.1: Given that $|\psi\rangle$ is a vector in Hilbert space, $\hat{A}$ must be (A). a vector, (B). a matrix, (C). a number.

Answer: The answer is (B). a matrix, since the operator $\hat{A}$ applied on the vector $|\psi\rangle$ produces another vector $|\phi\rangle$, $\hat{A}$ must be a matrix. Recall that an $n \times n$ matrix applied on an $n \times 1$ column vector produces another $n \times 1$ vector.

Linear operator: A linear operator acting on a linear superposition of state vectors results in

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a sum with the operator acting on each term individually,

$$\hat{A}\left(c_1|\psi_1\rangle + c_2|\psi_2\rangle\right) = c_1\hat{A}|\psi_1\rangle + c_2\hat{A}|\psi_2\rangle. \tag{3.3.2}$$

3.3.1 Properties of operators

1. **Equal operators:** Two operators $\hat{A}$ and $\hat{B}$ are equal if they produce the same state vector applied on arbitrary vector in the vector space $|\psi\rangle$,

$$\hat{A}|\psi\rangle = \hat{B}|\psi\rangle. \tag{3.3.3}$$

2. **Null operator:** Operator $\hat{A}$ is a null operator if it results in zero when acting on any state vector in the vector space,

$$\hat{A}|\psi\rangle = |0\rangle \equiv 0. \tag{3.3.4}$$

3. **Operator addition:** Given three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, addition of operators is associative,

$$\hat{A} + \left(\hat{B} + \hat{C}\right) = \left(\hat{A} + \hat{B}\right) + \hat{C} \tag{3.3.5}$$

and commutative,

$$\hat{A} + \hat{B} = \hat{B} + \hat{A}. \tag{3.3.6}$$

4. **Operator multiplication:** Given three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, multiplication of operators is associative,

$$\hat{A}\left(\hat{B}\hat{C}\right) = \left(\hat{A}\hat{B}\right)\hat{C}. \tag{3.3.7}$$

but not commutative, in general,

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}. \tag{3.3.8}$$

3.3.2 Hermitian conjugate (or Hermitian adjoint) operator in the dual space

Given operator $\hat{A}$, which acts on vector $|\psi\rangle$ in the vector space from the left as

$$\hat{A}|\psi\rangle, \tag{3.3.9}$$

the Hermitian adjoint operator $\hat{A}^\dagger$ acts on the dual vector $\langle\psi|$ from the right as

$$\langle\psi|\hat{A}^\dagger, \tag{3.3.10}$$

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where $\hat{A}^\dagger$ is defined as the transpose and complex conjugate of $\hat{A}$,

$$\hat{A}^\dagger = (\hat{A})^{T*} \tag{3.3.11}$$

There is a distinction in the interpretations of $\hat{A}|\psi\rangle$ and $\langle\psi|\hat{A}^\dagger$. The operator $\hat{A}$ in $\hat{A}|\psi\rangle$ acts to the right on $|\psi\rangle$ while the operator $\hat{A}^\dagger$ acts to the left on $\langle\psi|$.

The Hermitian conjugate of the vector $\hat{A}|\psi\rangle$ itself is then

$$\left(\hat{A}|\psi\rangle\right)^\dagger = |\psi\rangle^\dagger \hat{A}^\dagger = \langle\psi|\hat{A}^\dagger, \tag{3.3.12}$$

where

$$\langle\psi| = |\psi\rangle^\dagger, \tag{3.3.13}$$

Taking the Hermitian conjugate again, $\langle\psi|^\dagger = (|\psi\rangle^\dagger)^\dagger = |\psi\rangle$; i.e.,

$$\langle\psi|^\dagger = |\psi\rangle. \tag{3.3.14}$$

Thus, a bra vector is the Hermitian conjugate of a ket vector, and vice versa.

Note that for a product of operators, $(\hat{A}\hat{B})^\dagger = (\hat{A}\hat{B})^{T*} = (\hat{B}^T\hat{A}^T)^* = \hat{B}^{T*}\hat{B}^{T*} = \hat{B}^\dagger\hat{A}^\dagger$. Furthermore given $c\hat{A}$, where c is a complex number, we have $(c\hat{A})^\dagger = c^*\hat{A}^\dagger$.

3.3.3 Outer product operators

Given two state vectors $|\psi\rangle$ and $|\phi\rangle$, we can construct an operator using the outer product

$$|\psi\rangle\langle\phi|. \tag{3.3.15}$$

This operator can then act on a state vector $|\chi\rangle$ as

$$[|\psi\rangle\langle\phi|]\,|\chi\rangle = |\psi\rangle\langle\phi|\chi\rangle. \tag{3.3.16}$$

Taking the Hermitian conjugate,

$$(|\psi\rangle\langle\phi|\chi\rangle)^\dagger = \langle\chi|\phi\rangle\langle\psi|. \tag{3.3.17}$$

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On the other hand, noting that $\langle\phi|\chi\rangle$ is a complex number, we rewrite the Hermitian conjugate of (3.3.16) as

$$(|\psi\rangle\langle\phi|\chi\rangle)^\dagger = \langle\phi|\chi\rangle^*\langle\psi|. \tag{3.3.18}$$

Equating (3.3.17) and (3.3.18),

$$\langle\chi|\phi\rangle = \langle\phi|\chi\rangle^*. \tag{3.3.19}$$

This also follows directly noting that $\langle\phi|\chi\rangle$ is a complex number and taking the Hermitian conjugate is equivalent to taking the complex conjugate, since transposing has no effect on a number.

3.3.4 Matrix elements of operators

The matrix element of operator $\hat{A}$ between states $|\phi\rangle$ and $|\psi\rangle$ is defined as

$$\langle\phi|\hat{A}|\psi\rangle. \tag{3.3.20}$$

Because the matrix element is a complex number, its Hermitian conjugate is equivalent to its complex conjugate, and taking the Hermitian conjugate, we have

$$\langle\phi|\hat{A}|\psi\rangle^\dagger = \langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*, \tag{3.3.21}$$

or

$$\langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*, \tag{3.3.22}$$

which relates the matrix elements of $\hat{A}^\dagger$ and $\hat{A}$.

3.3.5 Hermitian operators

A Hermitian operator $\hat{A}$ is one for which the Hermitian conjugate is identical to the operator,

$$\hat{A}^\dagger = \hat{A}. \tag{3.3.23}$$

In other words, given $\hat{A}|\psi\rangle$ in the vector space, we have $\langle\psi|\hat{A}^\dagger = \langle\psi|\hat{A}$ in the dual vector space.

3.3.6 Matrix elements of a Hermitian operator

Question 3.3.2: For matrix elements of a Hermitian operator $\hat{A}$, (A). $\langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}|\phi\rangle$, (B). $\langle\phi|\hat{A}|\psi\rangle^* = \langle\phi|\hat{A}|\psi\rangle$, (C). $\langle\phi|\hat{A}|\phi\rangle = \langle\psi|\hat{A}|\psi\rangle$, (D). $\langle\phi|\hat{A}|\psi\rangle = \langle\phi|\hat{A}^\dagger|\psi\rangle$.

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Given a Hermitian operator $\hat{A}$, its matrix between states $|\phi\rangle$ and $|\psi\rangle$ satisfies, noting that $\hat{A}^\dagger = \hat{A}$ and that $\langle\phi|\hat{A}|\psi\rangle$ is a number , $\langle\phi|\hat{A}|\psi\rangle^\dagger = \langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\psi|\hat{A}|\phi\rangle$; i.e.,

$$\langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}|\phi\rangle. \tag{3.3.24}$$

In particular, taking $|\phi\rangle = |\psi\rangle$, the expectation value of $\hat{A}$ in state $|\psi\rangle$ satisfies

$$\langle\psi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}|\psi\rangle \equiv \langle\hat{A}\rangle. \tag{3.3.25}$$

In other words, a Hermitian operator has real expectation values. Observe that $\hat{A}^\dagger = \hat{A}$ also means that

$$\langle\phi|\hat{A}|\psi\rangle = \langle\phi|\hat{A}^\dagger|\psi\rangle \tag{3.3.26}$$

In other words, we get the same result whether $\hat{A}$ acts to the right or to the left.

3.3.7 Eigenvalues and eigenvector of Hermitian operators

If an operator $\hat{A}$ acting on a state vectors $|a_n\rangle$ produces a number a_n times the same vector as

$$\hat{A}|a_n\rangle = a_n|a_n\rangle \tag{3.3.27}$$

we say $|a_n\rangle$ is an eigenvector of $\hat{A}$ with eigenvalue a_n. For a Hermitian operator $\hat{A}$, the left hand side of (3.3.24) with $|\psi\rangle = |a_n\rangle$ and $|\phi\rangle = |a_m\rangle$ gives $\langle a_m|\hat{A}|a_n\rangle^* = a_n^*\langle a_m|a_n\rangle^* = a_n^*\langle a_n|a_m\rangle$, and the right hand gives $\langle a_n|\hat{A}|a_m\rangle = a_m\langle a_n|a_m\rangle$. Equating these,

$$(a_n^* - a_m)\,\langle a_n|a_m\rangle = 0. \tag{3.3.28}$$

Therefore, for $m = n$, we have $\langle a_n|a_n\rangle \neq 0$ and $a_n^* = a_n$. In other words, the eigenvalues are real.

For $m \neq n$, and assuming nondegenerate states, such that different eigenstates correspond to different eigenvalues, we have $a_n^* - a_m \neq 0$ and $\langle a_m|a_n\rangle = 0$. In other words, the eigenvectors are orthogonal. With normalized eigenvectors, we then have

$$\langle a_m|a_n\rangle = \delta_{mn}. \tag{3.3.29}$$

If the system has degenerate eigenvectors, different eigenvectors corresponding to the same eigenvalues, one uses the Gram-Schmidt orthogonalization to construct an orthonormal set of

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eigenvectors.

3.3.8 Completeness of eigenvectors of a Hermitian operator

Given the eigenvectors $|a_n\rangle$ of a Hermitian operator $\hat{A}$ for a system, can we express an arbitrary normalized state $|\psi\rangle$ of the system as a linear superposition of $|a_n\rangle$ as $|\psi\rangle = \sum_n c_n|a_n\rangle$? Is the set of $|a_n\rangle$ complete? Consider a normalized state $|\psi\rangle$ and define

$$|\Psi\rangle \equiv |\psi\rangle - \sum_n |a_n\rangle\langle a_n|\psi\rangle. \tag{3.3.30}$$

Because $|\psi\rangle$ is normalized,

$$\begin{aligned}
\langle\Psi|\Psi\rangle &= \left(\langle\psi| - \sum_m \langle\psi|a_m\rangle\langle a_m|\right)\left(|\psi\rangle - \sum_n |a_n\rangle\langle a_n|\psi\rangle\right) \\
&= \langle\psi|\psi\rangle - \sum_m \langle\psi|a_m\rangle\langle a_m|\psi\rangle - \sum_n \langle\psi|a_n\rangle\langle a_n|\psi\rangle + \sum_{nm} \langle\psi|a_m\rangle\delta_{mn}\langle a_n|\psi\rangle \\
&= \langle\psi|\psi\rangle - \sum_n \langle\psi|a_n\rangle\langle a_n|\psi\rangle \qquad (3.3.31) \\
&= \langle\psi|\psi\rangle - \sum_n |\langle a_n|\psi\rangle|^2 \qquad (3.3.32) \\
&= 1 - 1 = 0, \qquad (3.3.33)
\end{aligned}$$

since $\langle\psi|\psi\rangle = 1$ for a normalized $|\psi\rangle$ and $\sum_n |\langle a_n|\psi\rangle|^2 = 1$ for $|a_n\rangle$ containing all outcomes and summing over all n gives a total probability of one. Therefore, the set of $|a_n\rangle$ is complete if the sum of $|\langle a_n|\psi\rangle|^2$ equals one. In other words, if the state vector of every possible measurement outcome of $\hat{A}$ is included, since the total probability must be one. Looking at the next to last line in the equation above, it also means that

$$\sum_n |a_n\rangle\langle a_n| = 1; \tag{3.3.34}$$

i.e., $\sum_n |a_n\rangle\langle a_n|$ is an identity operator.

To clarify the notation, let us take the familiar example of a vector in 3-dimensions, $\vec{r} = (x, y, z)$. In our bra-ket notation, $|r\rangle = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x|1\rangle + x|2\rangle + x|3\rangle$, where $|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $|2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and

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$|3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Therefore, $x = \langle 1|r\rangle$, $y = \langle 2|r\rangle$, and $z = \langle 3|r\rangle$, or $|r\rangle = |1\rangle\langle 1|r\rangle + |2\rangle\langle 2|r\rangle + |3\rangle\langle 3|r\rangle$, where $|1\rangle\langle 1|r\rangle$ projects out the $x-$component together with its direction $|r\rangle$. In particular, $|1\rangle\langle 1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Therefore, $|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1$, which is the completeness relation, and $|r\rangle = |1\rangle\langle 1|r\rangle + |2\rangle\langle 2|r\rangle + |3\rangle\langle 3|r\rangle = (|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3|)\,|r\rangle = |r\rangle$. Observe that the completeness $\sum_n |n\rangle\langle n| = 1$ allows us to decompose and write any vector in 3-dimension in terms the basis vectors $|n\rangle$, $n = 1, 2, 3$.

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